# Homogenizer



Category: **Food Processing & Kitchen** Type: **Full SOP** Priority: **High** 2 labs

**Covers these lab items:** Homogenizer; Small Homogenizer

## What it is used for

A homogenizer reduces particle or droplet size and helps create a more uniform mixture in suitable food products or laboratory samples. Use the correct homogenizing attachment, speed, and operating procedure for the material being processed.

## Main parts

- Main motor and drive unit
- Homogenizing head, rotor, or probe
- Processing vessel or sample container
- Speed and power controls
- Shaft, coupling, and protective guard, if fitted
- Power cord and compatible accessories

## Before you start

1. Confirm that the homogenizer is suitable for the material and batch size.
2. Inspect the motor, probe, shaft, cable, and attachments for damage.
3. Clean and sanitise food-contact parts according to the hygiene procedure.
4. Check that the processing vessel is compatible and securely positioned.
5. Select the required speed and processing time according to the product procedure.
6. Wear suitable PPE and ensure the work area is stable and uncluttered.

## How to use

1. Switch off and disconnect the unit before fitting the homogenizing attachment.
2. Secure the correct probe or head according to the manufacturer's instructions.
3. Place the material in a compatible vessel without exceeding the recommended fill level.
4. Position the homogenizing head correctly within the material before starting.
5. Connect the unit to a suitable power supply and switch it on.
6. Start at the recommended low speed and increase gradually if required.
7. Maintain the required probe position and avoid contact with the vessel walls.
8. Monitor the mixture for uniformity, splashing, excessive heating, or unusual vibration.
9. Stop the unit before lifting the probe out of the material.
10. Switch off and disconnect the power before removing the attachment.
11. Transfer the processed material using clean, suitable equipment.
12. Clean and sanitise the probe and compatible parts after use.
13. Record processing settings and report any equipment faults.

## ⚠ Safety

- Only trained personnel should operate the homogenizer.
- Never touch a moving probe or rotor during operation.
- Keep hands, loose clothing, and utensils away from moving parts.
- Do not operate with a damaged shaft, loose attachment, or faulty electrical connection.
- Use suitable containment and PPE when processing hazardous or potentially infectious samples.
- Disconnect power before cleaning, changing attachments, or clearing a blockage.

## Care & storage

- Clean and sanitise product-contact parts after each batch.
- Prevent liquid from entering the motor housing or electrical connections.
- Inspect the shaft, coupling, seals, and probe for wear or damage.
- Store attachments clean and dry in a protected location.
- Arrange servicing according to the manufacturer's maintenance schedule.

## Common mistakes to avoid

- Starting the homogenizer before the probe is correctly positioned.
- Exceeding the recommended vessel fill level or batch capacity.
- Removing the probe while it is still rotating.
- Using incompatible cleaning chemicals or immersing non-waterproof parts.

*Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.*



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